Notes: Hai & Heckman (JPE, 2026)
The Causal Effects of Youth Cigarette Addiction and Education

Contents

1 Big picture **3**

2 Data and measurement **4**

2.1 Main data source: NLSY97 . . . 4

2.2 Smoking measurement . . . 4

2.3 Education measurement . . . 4

2.4 Ability and family background . . . 5

2.5 Cigarette prices and excise taxes . . . 5

2.6 Survival and mortality data . . . 5

2.7 Transfers, borrowing, and earnings . . . 5

3 Model structure and equations **7**

3.1 Choice set and life-cycle timing . . . 7

3.2 Addiction stock . . . 7

3.3 Endogenous survival probability . . . 7

3.4 Subjective discount factor . . . 7

3.5 Flow utility . . . 8

3.6 Dynamic optimization problem . . . 8

3.7 Budget and borrowing constraints . . . 8

3.8 Earnings function . . . 9

3.9 Measurement system for unobserved abilities . . . 10

4 Identification and estimation strategy **12**

4.1 What is identified structurally? . . . 12

4.2 Ability measurement parameters . . . 12

4.3 Smoking-specific parameters . . . 12

4.4 Subjective discount factor parameters . . . 13

4.5 Schooling-specific parameters . . . 13

4.6 Risk aversion and earnings . . . 13

4.7 Two-step estimation . . . 13

5 Estimation results and model fit **15**

5.1 Raw empirical gradients: Figure 1 and Table 1 . . . 15

5.2 Smoking initiation: Figure 2 . . . 15

5.3 Key parameter estimates: Table 2 . . . 16
5.4 Model fit through targeted regression moments: Table 3 . . . 16
5.5 Life-cycle and nontargeted fit: Figures 3 and 4 . . . 17
5.6 Longevity implications: Table 4 . . . 18

6 Policy simulations and causal counterfactuals **19**
6.1 Education to smoking: one extra year of schooling . . . 19
6.2 Who quits when education increases? Table 5 . . . 19
6.3 Smoking to education: eliminating smoking . . . 20
6.4 Corrective excise taxes . . . 20
6.5 Distributional effects of the tax . . . 21
6.6 Alternative transfer schemes . . . 21

7 Limitations and interpretation **23**
7.1 Structural assumptions . . . 23
7.2 Peer effects . . . 23
7.3 Substitution into other risky behaviors . . . 23
7.4 Policy interpretation . . . 23

8 Tables and figures **24**
8.1 Figure 1, Table 1, and Figure 2: empirical starting point . . . 24
8.2 Key model equations . . . 25
8.3 Estimation and fit . . . 26
8.4 Counterfactuals and policy . . . 27

9 Short summary **29**

10 Conclusion **29**
10.1 Core conclusion . . . 29
10.2 Economic intuition . . . 29
10.3 Policy takeaway . . . 30
10.4 What to remember . . . 30

1 Big picture

- **Core question.** The paper asks two reciprocal causal questions: does youth cigarette addiction reduce educational attainment, and does education reduce cigarette addiction?
- **Why it is difficult.** Smoking and schooling are jointly determined. Observable and unobservable traits, especially cognitive ability, noncognitive ability, parental background, patience, and peer environments, can jointly predict both outcomes. A reduced-form comparison of smokers and nonsmokers is therefore not enough.
- **Method.** The authors develop and estimate a forward-looking life-cycle model in which young men choose smoking, schooling, part-time work, consumption, saving, and borrowing under borrowing constraints.
- **Central state variable.** Smoking builds an addiction stock. The addiction stock affects the future utility of smoking, subjective discounting, mortality risk, schooling utility, and borrowing limits.
- **Two-way mechanism.** Smoking lowers schooling incentives through greater myopia, lower schooling utility, cigarette costs, and lower expected longevity. Education lowers smoking through lower smoking utility, greater patience, higher survival, higher income, and better health knowledge.
- **Identification.** The paper uses variation in initial endowments, the measurement system for cognitive/noncognitive abilities, state-year cigarette prices and excise taxes, net worth moments, earnings moments, and dynamic life-cycle moments.
- **Data.** The main sample is male NLSY97 respondents born in 1982 or later, observed over 1997–2015. The final estimation sample contains 1,605 individuals ages 15–33.
- **Main education-to-smoking result.** A counterfactual increase of one extra year of schooling at age 20 reduces the smoking rate by 3 percentage points at age 20 and by roughly 7 percentage points by age 30.
- **Main smoking-to-education result.** Eliminating smoking as a choice raises college attendance by roughly 2 percentage points in the population and by about 5 percentage points among benchmark smokers.
- **Tax policy result.** A revenue-neutral 25% additional cigarette excise tax, with proceeds rebated as tuition subsidies, produces an education effect close to the no-smoking counterfactual and lowers the age-30 daily smoking rate from 24.48 percentage points to 22.66 percentage points.
- **Heterogeneity.** Effects vary strongly across cognitive and noncognitive ability. The smoking-to-education margin is especially important for youth with median cognitive ability and low noncognitive ability.
- **Takeaway.** Youth smoking is not only a future health risk. It can reduce human capital investment during adolescence and early adulthood. Policies that reduce youth smoking can also raise education.

Interpretation. The paper is a structural causal exercise, not a simple natural experiment. The credibility of the counterfactuals comes from combining economic structure with targeted moments that discipline smoking, schooling, wealth, prices, earnings, ability, and survival.

2 Data and measurement

2.1 Main data source: NLSY97

- The main dataset is the National Longitudinal Survey of Youth 1997 (NLSY97), which follows youth born between 1980 and 1984.
- The paper focuses on males in the core cross-sectional sample to avoid confounding by fertility choices and because the core sample does not require oversampling weights.
- The sample is restricted to individuals born in 1982 or later so that respondents are age 15 or younger in the initial 1997 interview. This allows the model to start before most regular daily smoking begins.
- The final sample is unbalanced and contains 1,605 individuals from ages 15 to 33.
- The NLSY97 observes smoking, education, employment, earnings, parental transfers, government transfers, parental education, and ability measures.

Interpretation. This dataset is well matched to the question because it observes early-life choices annually. Since regular smoking initiation is concentrated before age 21, observing people from age 15 is crucial.

2.2 Smoking measurement

- Smoking is measured as daily or regular smoking, not merely having tried cigarettes.
- The model distinguishes current smoking, past smoking, accumulated years smoked, and addiction stock.
- The empirical facts show that smoking is concentrated early: around 15% of eventual smokers initiate regular smoking at age 15; initiation peaks around ages 16–17; by age 21, about 86% of eventual regular smokers have already initiated.

Interpretation. The paper focuses on addiction-relevant smoking, not experimentation. This matters because the model treats smoking as a dynamic stock-building activity.

2.3 Education measurement

- Education is measured as years of completed schooling.
- College attendance is measured by whether an individual attends college by age 30.
- Schooling enrollment is modeled as a yearly discrete choice until schooling choices end.
- In the raw data, high-school-or-less individuals have a smoking rate of 0.30 at age 30, whereas college-and-more individuals have a smoking rate of 0.10.
- In the other direction, youth who smoked before age 20 have 12.13 years of schooling at age 30, whereas those who did not smoke before age 20 have 14.34 years.

Interpretation. The raw education-smoking gradient is large in both directions, but it is not causal by itself. The structural model asks how much of this association reflects reciprocal causal effects rather than selection.

2.4 Ability and family background

- Cognitive ability is measured using four ASVAB components: arithmetic reasoning, mathematics knowledge, paragraph comprehension, and word knowledge.
- Noncognitive ability is measured using six behavior and socioemotional measures, including early adverse behavior and behavioral/emotional functioning indicators.
- Parental education affects initial ability distributions, parental transfers, and schooling preferences.
- The model treats cognitive and noncognitive abilities as latent factors measured with error.

Interpretation. Ability is central because it can generate selection into both schooling and smoking. If it were ignored, estimated causal effects would mix true effects with pre-existing differences in intelligence, self-control, and family background.

2.5 Cigarette prices and excise taxes

- The authors obtain average costs and excise taxes per pack of cigarettes from the Centers for Disease Control and Prevention.
- Cigarette price data are merged to NLSY97 respondents by state of residence and year.
- State-year price variation helps identify smoking-related parameters.
- In the identification strategy, past and future cigarette prices and excise taxes are used as instruments in auxiliary regressions.

Interpretation. Cigarette prices provide an external source of variation in smoking costs. This is important for separating causal effects of smoking from unobserved propensities to smoke.

2.6 Survival and mortality data

- Survival probabilities are not taken directly from aggregate life tables because the model requires survival to depend on smoking and education.
- For older ages, the paper estimates survival using the Health and Retirement Study.
- For younger ages, it combines mortality multiple-cause-of-death data with population data from the US census.
- Survival depends on age, current smoking, past smoking, and education.

Interpretation. Endogenous survival is not an auxiliary detail. It affects the value of future schooling returns, the desire to smoke, and the natural borrowing limit.

2.7 Transfers, borrowing, and earnings

- Parental transfers include allowances, nonallowance transfers, financial aid gifts, inheritance, and other monetary help.
- Government transfers are modeled as a means-tested consumption subsidy that helps maintain a consumption floor.
- Agents can borrow up to natural borrowing limits, which depend on education, earnings potential, survival, and smoking.

- Earnings are modeled with an augmented Mincer function depending on education, potential experience, age, cognitive ability, noncognitive ability, and earnings shocks.

Interpretation. Borrowing constraints make smoking and schooling interact through the budget. Cigarette purchases can crowd out school costs, while education can expand future borrowing capacity by increasing earnings and survival.

3 Model structure and equations

3.1 Choice set and life-cycle timing

- Agents are economically active from ages 15 to 64.
- Each year, they choose consumption, next-period savings, whether to smoke, whether to attend school, and whether to work part-time while in school.
- Schooling choices end before age 29; high school choices end before age 21.
- From age 65 onward, agents retire and choose only consumption and savings.
- Agents are forward-looking and understand the future consequences of current smoking and schooling decisions.

Interpretation. Starting the model at age 15 aligns with the data and with the onset of regular smoking. Ending schooling choices before age 29 makes the schooling decision a finite-horizon investment problem.

3.2 Addiction stock

The addiction stock evolves according to

$$q_t = (1 - \delta_q) q_{t-1} + d_{q,t-1}, \qquad t \geq 1, \tag{1}$$

where $d_{q,t-1}$ is the previous period smoking decision and δ_q is the depreciation rate of addiction.

Interpretation. The stock formulation makes smoking intertemporally nonseparable. Smoking today raises tomorrow's addiction stock, which changes tomorrow's smoking utility, discounting, schooling utility, and survival-related incentives.

$$q_t = (1 - \delta_q) q_{t-1} + d_{q,t-1}, \quad t \geq 1, \tag{1}$$

3.3 Endogenous survival probability

The one-period survival probability is written conceptually as

$$p_t = p_t(d_{q,t}, q_t, d_{e,t} + e_t), \tag{2}$$

where survival depends on current smoking, past smoking/addiction, current education, and age.

Interpretation. Education and smoking affect the effective horizon over which agents value future outcomes. Smoking shortens the perceived future, while education lengthens it through better survival.

3.4 Subjective discount factor

The subjective discount factor depends on addiction and ability:

$$\beta(q_t, v) = \exp(-k_q q_t) \exp(-k_r + k_c v_c + k_n v_n). \tag{3}$$

- $k_q \geq 0$ captures the myopia effect of addiction.
- k_r controls the baseline subjective discount rate.
- k_c and k_n capture how cognitive and noncognitive ability shift patience.

Interpretation. This is one of the paper's main innovations. Time preference is not an exogenous type. It changes with smoking addiction and differs with ability endowments.

$$\beta(q_t, \boldsymbol{\theta}) \coloneqq \underbrace{\exp(-\kappa_q q_t)}_{\text{effects of addiction}} \underbrace{\exp(-\kappa_r + \kappa_c \theta_c + \kappa_n \theta_n)}_{\text{effects of individual traits}}, \tag{2}$$

arameter $\kappa_q \geq 0$ captures the impact of addiction capital on
iscount factors and hence on time discount rates. As addic-

3.5 Flow utility

The period utility function has CRRA consumption utility plus smoking and schooling components:

$$U(c_t, d_t; q_t, e_t, v, X_t, \epsilon_t) = \frac{c_t^{1-\gamma} - 1}{1-\gamma} + u_q(d_{q,t}; q_t, e_t, v, \epsilon_{q,t}) + u_e(d_{e,t}, d_{p,t}; q_t, X_t, v, \epsilon_{e,t}). \tag{4}$$

The smoking utility includes education, ability, preference shocks, addiction stock, and an addiction-stock-squared term. The schooling utility includes addiction stock, part-time work, ability, controls, and schooling shocks.

Interpretation. The signs are economically meaningful. Education lowers smoking utility, addiction lowers schooling utility, and ability raises schooling utility. This generates dynamic substitution between smoking and schooling.

$$\begin{aligned}
U(c_t, \boldsymbol{d}_t; q_t, e_t, \boldsymbol{\theta}, \boldsymbol{X}_t, \boldsymbol{\epsilon}_t) &= \frac{c_t^{1-\gamma} - 1}{1-\gamma} + \underbrace{u_q(d_{q,t}; q_t, e_t, \boldsymbol{\theta}, \boldsymbol{\epsilon}_{q,t})}_{\text{smoking utility}} + \underbrace{u_e(d_{e,t}, d_{p,t}; q_t, \boldsymbol{X}_t, \boldsymbol{\theta}, \boldsymbol{\epsilon}_{e,t})}_{\text{schooling utility}}, \\
u_q(\cdot) &= d_{q,t} \cdot (\boldsymbol{\nu}_e e_t + \boldsymbol{\nu}_\theta \boldsymbol{\theta} + \nu_0 + \boldsymbol{\epsilon}_{q,t}) + \nu_1 q_t + \nu_q q_t d_{q,t} + \nu_{q^2} q_t^2 \\
&= d_{q,t} \cdot (\boldsymbol{\nu}_e e_t + \boldsymbol{\nu}_\theta \boldsymbol{\theta} + \nu_0 + \boldsymbol{\epsilon}_{q,t}) + \underbrace{(\nu_1 + \nu_q) q_t d_{q,t}}_{\text{addictive nature of smoking}} \\
&\quad + \nu_1 q_t (1 - d_{q,t}) + \underbrace{\nu_{q^2} q_t^2}_{\text{negative health effect}}, \\
u_e(\cdot) &= d_{e,t} \cdot (\xi_q q_t + \xi_p d_{p,t} + \boldsymbol{\xi}_\theta \boldsymbol{\theta} + \boldsymbol{\xi}_e \boldsymbol{X}_t + \boldsymbol{\epsilon}_{e,t}),
\end{aligned} \tag{3}$$

3.6 Dynamic optimization problem

The value function is recursively defined as

$$V_t(\Omega_t) = \max_{d_t, c_t, s_{t+1}} \{U(c_t, d_t; q_t, e_t, v, X_t, \epsilon_t) + \beta(q_t, v) p_t(d_{q,t}, q_t, d_{e,t} + e_t) E_t[V_{t+1}(\Omega_{t+1}) \mid \Omega_t]\}. \tag{5}$$

Interpretation. Survival probability and subjective discounting both multiply the continuation value. Therefore, smoking can lower education by reducing the value of future payoffs through two channels: physiological mortality risk and subjective impatience.

3.7 Budget and borrowing constraints

The budget constraint is

$$s_{t+1} - s_t(1 + r(s_t)) = Y_t + T_t + G_t - ig(c_t + p_t d_{q,t} + me_{t+1} d_{e,t}), \tag{6}$$

and the borrowing constraint is

$$s_{t+1} \geq -\bar{S}_{t+1}(d_{q,t}, q_t, e_{t+1}, v). \tag{7}$$

Interpretation. Cigarettes and education both use current resources. However, education raises future earnings and survival, while smoking raises mortality and can tighten borrowing limits. This creates a financially meaningful smoking-schooling tradeoff.

$$V_t(\Omega_t) = \max_{\{\boldsymbol{d}_t, c_t, s_{t+1}\}} \{U(c_t, \boldsymbol{d}_t; q_t, e_t, \boldsymbol{\theta}, \boldsymbol{X}_t, \boldsymbol{\epsilon}_t) + \beta(q_t, \boldsymbol{\theta})\pi_t(d_{q,t}, q_t, d_{e,t} + e_t) \cdot \mathbb{E}_t[V_{t+1}(\Omega_{t+1})|_{\Omega_t}]\}.$$

After age 64, remaining lifetime utility becomes

$$V_{65}(\Omega_{65}) = \max_{\{c_t\}} \sum_{t=65}^{95} \left\{ \left(\prod_{j=65}^{t-1} \pi_j(d_{q,64}, q_j, e_{64})\beta(q_j, \boldsymbol{\theta}) \right) \left(\frac{c_t^{1-\gamma} - 1}{1-\gamma} + u_q(d_{q,64}; q_t, e_{64}, \boldsymbol{\theta}, 0) \right) \right\}.$$

The individual optimization is subject to the budget constraint, the borrowing constraint, and the law of motion of addiction stock q (eq. [1]). An individual discounts the future using his endogenous survival probability $\pi_t(\cdot)$ and subjective discount factor $\beta(\cdot)$ (eq. [2]).

2. Budget Constraint and Borrowing Constraint

An individual's budget constraint and borrowing constraint are, respectively,[9]

$$\underbrace{s_{t+1} - s_t \cdot (1 + r(s_t))}_{\text{savings in period } t} = \underbrace{Y_t + T_t + G_t}_{\text{earnings, transfers, and subsidies}} - \underbrace{(c_t + p_t \cdot d_{q,t} + m_{e_{t+1}} \cdot d_{e,t})}_{\text{consumption, addiction, and education costs}} \tag{4}$$

and

$$s_{t+1} \geq \underline{S}_{t+1}(d_{q,t}, q_t, e_{t+1}, \theta), \tag{5}$$

3.8 Earnings function

The full-time earnings function is an augmented Mincer specification:

$$\ln Y(e_t, k_t, t, v, \epsilon_{y,t}) = q_0 + q_{k,1}k_t + q_{k,2}k_t^2 + q_{k,3}k_t 1(e_t \geq 16) + q_c v_c + q_n v_n + exteducationterms + \text{age terms} + \epsilon_{y,t}. \tag{8}$$

Interpretation. Education affects smoking partly because schooling raises future income. Higher future earnings increase the return to schooling and can change the opportunity cost of addiction.

$$
\begin{aligned}
\ln Y(e_t, k_t, t, \boldsymbol{\theta}, \epsilon_{y,t}) \;=\; & \omega_0 + \omega_{k,1} k_t + \omega_{k,2} k_t^2 + \underbrace{\omega_{k,3} k_t \mathbf{1}(e_t \geq 16)}_{\text{higher return to experience by education}} + \underbrace{\omega_c\, \theta_c + \omega_n \theta_n}_{\text{reward to ability}} \\
& + \underbrace{(\omega_{e,1} \mathbf{1}(e_t \geq 12) + \omega_{e,2} \mathbf{1}(e_t < 12))(e_t - 12) + \omega_{e,3} \mathbf{1}(12 \leq e_t < 16) + (\omega_{e,4} + \omega_{e,5}(e_t - 16)) \mathbf{1}(e_t \geq 16)}_{\text{effect by education level}} \\
& + \underbrace{\omega_{a,1}(18 - t)\mathbf{1}(t < 18) + \omega_{a,2}(20 - t)\mathbf{1}(18 \leq t \leq 20)}_{\text{age effect}} + \epsilon_{y,t},
\end{aligned}
\tag{7}
$$

where $k_t \geq 0$ is the number of years of potential experience and $\epsilon_{y,t} \geq \underline{\epsilon}_y$ is

3.9 Measurement system for unobserved abilities

For each measure j of ability type $k \in \{c, n\}$, the latent measurement equation is

$$
Z^*_{k,j} = \mu_{z,k,j} + \alpha_{z,k,j} v_k + x'_z \beta_{z,k,j} + \epsilon_{z,k,j}. \tag{9}
$$

Observed measures are continuous, binary, or ordered discrete functions of $Z^*_{k,j}$. The latent abilities are jointly normal:

$$
\begin{pmatrix} v_c \\ v_n \end{pmatrix} \sim N\left(\begin{pmatrix} \mu_c(e^{pr}) \\ \mu_n(e^{pr}) \end{pmatrix}, \begin{pmatrix} \sigma_c^2 & \sigma_{cn} \\ \sigma_{cn} & \sigma_n^2 \end{pmatrix} \right). \tag{10}
$$

Interpretation. This measurement system is important because ability is not directly observed. It lets the authors use multiple proxies to infer latent cognitive and noncognitive endowments and then carry those latent traits into the structural model.

$$Z^*_{k,j} = \mu_{z,k,j} + \alpha_{z,k,j}\theta_k + \boldsymbol{x}'_z\boldsymbol{\beta}_{z,k,j} + \epsilon_{z,k,j}, \tag{8}$$

$$Z_{k,j} = \begin{cases} Z^*_{k,j} & \text{if } Z_{k,j} \text{ is continuous,} \\ \mathbf{1}(Z^*_{k,j} > 0) & \text{if } Z_{k,j} \text{ is an indicator variable,} \\ \mathbf{1}(Z^*_{k,j} > 0) + \mathbf{1}(Z^*_{k,j} > \text{cutoff}_{k,j}) & \text{if } Z_{k,j} \text{ takes values } \{0, 1, 2\}, \end{cases} \tag{9}$$

where $\boldsymbol{x}_z$ is a vector of individual control variables observed in the initial survey year 1997, including parental education, youth age in 1997, and youth schooling level in 1997. The measurement errors are assumed to be independently distributed across equations and over time. We also assume that the measurement errors follow normal distributions: $\epsilon_{z,k,j} \sim N(0, \sigma^2_{z,k,j})$ for $k \in \{c, n\}$ and $j \in \{1, \ldots, J_k\}$.

We assume that the unobserved endowments in cognitive and noncognitive abilities are jointly normally distributed and independent of all shocks specified in the model $\boldsymbol{\epsilon}_t$ at a point in time and over time. These skill endowments are either inherited from parents or formed as a result of parental investment, due to parents' knowledge, preferences, or financial resources. The means of initial endowments may differ depending on parental education e_{pr}—that is, $\mu_k(e_{\text{pr}}) = \mu_{k,0} + \mu_{k,1}e_{\text{pr}}$, for $k = c, n$. In particular, the ex ante joint distribution of unobserved cognitive and noncognitive abilities is assumed to be

$$\begin{pmatrix} \theta_c \\ \theta_n \end{pmatrix} \sim N\left(\begin{pmatrix} \mu_c(e_{\text{pr}}) \\ \mu_n(e_{\text{pr}}) \end{pmatrix}, \begin{pmatrix} \sigma^2_c & \sigma_{c,n} \\ \sigma_{c,n} & \sigma^2_n \end{pmatrix}\right), \tag{10}$$

4 Identification and estimation strategy

4.1 What is identified structurally?

The model estimates parameters governing:

- the addiction stock depreciation rate;
- subjective discounting and myopia;
- smoking utility;
- schooling utility;
- risk aversion;
- earnings;
- measurement of cognitive and noncognitive ability.

Interpretation. Identification relies on the fact that different parameters affect different margins: smoking dynamics, schooling enrollment, wealth, earnings, ability measurements, and price responsiveness.

4.2 Ability measurement parameters

- The factor model is estimated first using the observed cognitive and noncognitive measures.
- Continuous measures, binary measures, and ordered discrete measures are linked to latent abilities using linear, probit, and ordered probit measurement equations.
- Parental education shifts the mean of the latent ability distribution.

Interpretation. The first step separates latent traits from measurement error. This is essential because the structural model uses ability as an initial condition, not as an observed variable.

4.3 Smoking-specific parameters

- Smoking preference parameters are identified using smoking choices by first-time smokers and continuing smokers with different years smoked.
- The addiction depreciation rate is informed by how the response to cigarette prices depends on past smoking.
- Auxiliary 2SLS regressions instrument current, lagged, and future smoking with state-year cigarette price and excise tax variation.
- The cigarette-price effect reveals the monetary value of adding to addiction stock and therefore helps discipline dynamic addiction parameters.

Interpretation. The logic resembles rational-addiction identification: addiction is visible through dynamic responses to prices and through dependence of current smoking on past and future smoking.

4.4 Subjective discount factor parameters

- The authors use the distribution of net worth to identify discounting parameters.
- The correlation between net worth and accumulated smoking helps identify the myopia parameter k_q.
- The correlation between net worth and cognitive/noncognitive ability helps identify ability effects on patience.
- The median net worth level helps identify the baseline discounting parameter.

Interpretation. More impatient people save less. If addiction raises impatience, accumulated years smoked should be informative about net worth, conditional on ability and other states.

4.5 Schooling-specific parameters

- The key schooling auxiliary regression relates current enrollment to past smoking, cognitive ability, noncognitive ability, age, and education level.
- Past smoking is instrumented using lagged state-year cigarette prices.
- The coefficient on past smoking helps identify the effect of addiction on schooling utility.
- Remaining schooling preference parameters are identified from variation in schooling probabilities by age, education, ability, and parental background.

Interpretation. The authors use price-induced smoking variation to discipline the reverse channel from smoking into schooling.

4.6 Risk aversion and earnings

- Risk aversion is linked to the wealth distribution because borrowing constraints and uninsured earnings risk create precautionary saving incentives.
- The Gini coefficient of net worth is used as a targeted moment.
- Earnings function parameters are identified using earnings variation by education, experience, age, and latent ability scores.

Interpretation. Wealth inequality is not only an outcome to fit. It is used as evidence about risk preferences and dynamic saving incentives.

4.7 Two-step estimation

The first step estimates the ability measurement system by maximum likelihood:

$$\max \prod_i \int f(Z_{i,c}, Z_{i,n}; x_{i,z}, v_c, v_n) dF(v_c, v_n). \tag{11}$$

The second step estimates structural parameters by simulated method of moments:

$$J = \min_\theta \left(\widehat{M}_N - M_{\widetilde{N}}(\theta)\right)' \widehat{W}_N \left(\widehat{M}_N - M_{\widetilde{N}}(\theta)\right). \tag{12}$$

- The authors simulate 30 copies of each of the 1,605 NLSY97 individuals.

- The second-step simulated sample therefore has 48,150 individuals and 770,400 observations from ages 15 to 30.
- The second step estimates 43 parameters.
- There are 107 targeted empirical moments, including life-cycle moments and regression coefficients.

Interpretation. The estimation strategy is deliberately high dimensional. The model is not disciplined by one moment; it must jointly match smoking, schooling, earnings, net worth, prices, ability, and dynamic behavioral patterns.

5 Estimation results and model fit

5.1 Raw empirical gradients: Figure 1 and Table 1

Read.

- Figure 1A shows that earlier smoking initiation is associated with fewer years of schooling.
- Figure 1B shows that more education is associated with a lower age-30 smoking rate.
- Table 1A shows that age-30 smoking is 0.30 among high-school-or-less men and 0.10 among college-and-more men.
- Table 1B shows that men who smoked before age 20 have 12.13 years of schooling at age 30, compared with 14.34 for those who did not smoke before age 20.
- Table 1C shows life-cycle evolution: by age 20, the average number of years smoked is 0.71, and average schooling is 12.21 years.

Economic message. The descriptive evidence establishes a strong two-way education-smoking relationship. The structural analysis is needed because these gradients could be driven by unobserved ability, patience, and family background.

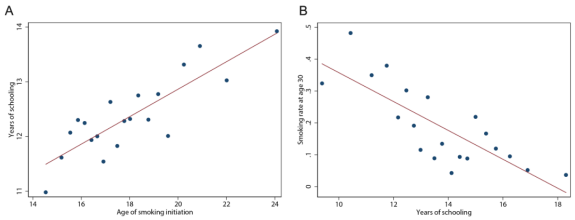

Fig. 1.—Empirical relationship between smoking and education. *A*, Binned scatterplot of the mean years of schooling as a function of age of smoking initiation among smokers. *B*, Binned scatterplot of the age 30 smoking rate as a function of years of schooling. Both plots control for cognitive and noncognitive ability average measures and parental college education. Years of schooling are measured at age 30. Source: NLSY97 males.

Figure 1: empirical relationship between smoking and education.

TABLE 1
Summary Statistics of Key Variables

	(1)	(2)	(3)	(4)
	A. Youth Smoking by Education (Age 30)			
	High School or Less	College and More	Difference	p-value of Difference
Smoking	.30	.10	−.20	.00
Years smoked	4.88	1.64	−3.24	.00
Never smoked	.39	.70	.31	.00
	B. Age 30 Education by Youth Smoking History			
	Smoked before Age 20	Never Smoked before Age 20	Difference	p-Value of Difference
Years of schooling (age 30)	12.13	14.34	2.21	.00
College and more (age 30)	.34	.68	.34	.00
	C. Key Variables by Age (NLSY97)			
	Age 15	Age 20	Age 25	Age 30
Years smoked as of $t-1$	.00	.71	1.80	2.90
Never smoked as of $t-1$	1.00	.71	.63	.58
School enrollment	.97	.35	.09	.00
Working part-time while in school	.13	.26	.06	.00
Years of schooling	8.46	12.21	13.36	13.60
Earnings after leaving school ($)	2,488	17,244	29,957	40,570

Table 1: summary statistics of key variables.

5.2 Smoking initiation: Figure 2

Read.

- Figure 2 shows the age distribution of smoking initiation among regular smokers.
- Initiation is concentrated in adolescence.
- Approximately 15% of smokers initiate at age 15.
- Initiation is highest around ages 16–17.
- By age 21, roughly 86% of smokers have already initiated.

Economic message. The policy-relevant stage is adolescence and early adulthood. Interventions that reduce smoking initiation can affect education before educational choices are completed.

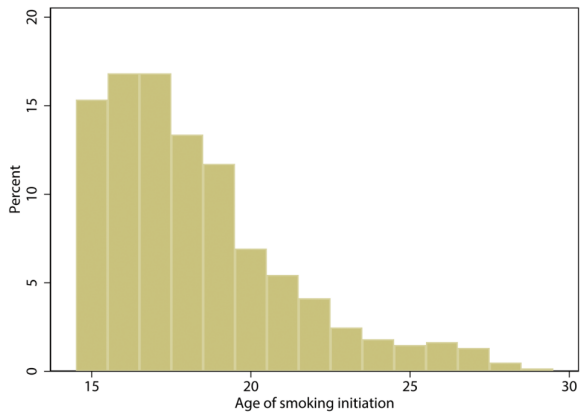

Fig. 2.—Age distribution of smoking initiation (among smokers).

TABLE 2
Key Parameter Estimates

Description	Symbol	Estimate	Standard Error
Depreciation rate of addiction stock	δ_q	.3684	.0186
Subjective discount rate terms:			
Subjective discounting rate	κ_r	.0213	.0009
Myopia effect of addiction stock	κ_q	.00046	.000024
Effect of cognitive ability	κ_c	.0033	.0008
Effect of noncognitive ability	κ_n	.0032	.0014
Addiction utility parameters:			
Addiction stock	ν_q	.0224	.0014
Addiction stock squared	ν_{q^2}	−.0017	.0012
Years of schooling −9	ν_e	−.0049	.0014
Cognitive ability	ν_c	−.00001	.0022
Noncognitive ability	ν_n	−.00785	.0024
Intercept	ν_0	.0075	.0029
SD of preference shock to smoking	σ_q	.0241	.0030
Schooling utility parameters:			
Effect of addiction on schooling utility	ξ_q	−.0011	.0043
Working part-time while in school	ξ_p	−.0141	.0026
Cognitive ability	$\xi_{\theta,c}$	.0598	.0088
Noncognitive ability	$\xi_{\theta,n}$	.0380	.0132
SD of preference shock to schooling	σ_e	.2353	.0145

Note.—Depreciation rate δ_q of addiction stock is from eq. (1): $q_t = (1-\delta_q)q_{t-1} + d_{q,t-1}$. Subjective discount factor terms are presented in eq. (2): $\beta(q,\theta) := \exp(-\kappa_q q)\exp(-\kappa_r + \kappa_c\theta_c + \kappa_n\theta_n)$. Utility preference terms are presented in eq. (3). In particular, smoking addiction preference terms are $u_q(\cdot) = d_{q,t}\cdot(\nu_0 + \nu_1 q_t + \nu_e e_t + \nu_\theta\theta + \epsilon_{q,t}) + \nu_2 q_t^2$. Schooling preferences terms are $u_e(\cdot) = d_{e,t}\cdot(\xi_q q_t + \xi_p d_{p,t} + \xi_\theta\theta + \xi_e X_t + \varepsilon_{e,t})$. The parameter estimates of ξ_e are reported in table C.VIII and discussed in sec. IV.A.3.

5.3 Key parameter estimates: Table 2

Read.

- The estimated addiction-stock depreciation rate is 0.3684, so addiction stock is persistent but depreciates over time.
- The baseline subjective discount-rate term is 0.0213.
- The myopia effect of addiction stock is 0.00046, meaning addiction slightly lowers the subjective discount factor per unit, but the effect accumulates dynamically.
- Cognitive and noncognitive ability raise subjective patience: estimated effects are 0.0033 and 0.0032.
- Education lowers smoking utility: the schooling coefficient in smoking utility is negative.
- Addiction utility initially rises with addiction stock but has a negative squared term, so the benefits of additional addiction eventually turn down.
- Cognitive and noncognitive ability raise schooling utility; working while in school lowers schooling utility.

Economic message. The parameter estimates are consistent with the paper's mechanism: smoking creates myopia and changes future utility; education and abilities make smoking less attractive and schooling more attractive.

5.4 Model fit through targeted regression moments: Table 3

Read.

- In the smoking participation regression, the model matches the negative effect of cigarette costs and the negative effect of education on smoking.
- The model also captures the strong state dependence in smoking, though some dynamic coefficients differ from the data.
- In the schooling attendance regression, past smoking has a negative effect on current enrollment in both the data and the model.
- In the net worth regression, the model closely matches the positive roles of cognitive ability and noncognitive ability and the negative relation between past smoking and wealth.

Economic message. The model fit matters because the counterfactuals rely on the model's ability to reproduce the causal margins that connect smoking, schooling, prices, and wealth.

TABLE 3
Model Fit: Coefficients of Regressions for Smoking, Schooling, and Net Worth

	Data	Model	SE of Data	| Data-Model | / SE of Data
	A. Smoking Participation 2SLS Regression			
Smoking at $t-1$ ($d_{g,t-1}$)	.5072	.6687	.1494	1.0811
Smoking at $t+1$ ($d_{g,t+1}$)	.4043	.2498	.1752	.8816
Years of schooling	−.0060	−.0075	.0063	.2403
Cognitive ability	.0025	−.0047	.0052	1.3668
Noncognitive ability	−.0029	−.0292	.0092	2.8391
Log cigarette cost (log $p_{g,t}$)	−.0126	−.0138	.0183	.0650
	B. Schooling Attendance 2SLS Regression			
Smoking at $t-1$ ($d_{g,t-1}$)	−.7376	−.4618	.4233	.6517
Cognitive ability	.1277	.1423	.0135	1.0790
Noncognitive ability	.0237	.0817	.0374	1.5543
	C. Log Net Worth OLS Regression			
Cognitive ability	.2882	.2874	.1320	.0059
Noncognitive ability	.0786	.1526	.1796	.4120
Years smoked as of $t-1$	−.0525	−.0198	.0206	1.5869

Note.—In panel A, the dependent variable is the indicator variable of smoking in current period ($d_{g,t}$). Past and future smoking participation decisions ($d_{g,t-1}$, $d_{g,t+1}$) are instrumented using the lags and leads of excise taxes and cigarette prices. The regression analysis is restricted to postschooling observations between ages 18 and 30. In panel B, the dependent variable is the indicator variable of schooling enrollment in the current period ($d_{e,t}$). The past smoking participation decision $d_{g,t-1}$ is instrumented using past cigarette price. The regression also controls for the indicator variables for age and education. The population of the regression analysis includes individuals aged 18–24 who were previously in school. In panel C, the dependent variable is the logarithm of net worth at age 30 for individuals with at least $100. In all three panels, cognitive ability and noncognitive ability factor scores are obtained from the first-stage estimation. Parameter estimates of the constant terms are not targeted moments and hence are not reported here.

5.5 Life-cycle and nontargeted fit: Figures 3 and 4

Read.

- Figure 3 shows the model fit to life-cycle moments, including smoking, schooling, and related outcomes over age.
- Figure 4A checks a nontargeted moment: college-aged enrolled youth are less likely to initiate smoking than nonenrolled youth.
- Figure 4B checks another nontargeted moment: relapse probability rises with years smoked among people who quit between ages 25 and 29.
- The nontargeted fits are important because they test model predictions not directly imposed in estimation.

Economic message. The model is not just interpolating targeted moments. It can also reproduce economically meaningful patterns in smoking initiation and relapse that validate the addiction-state mechanism.

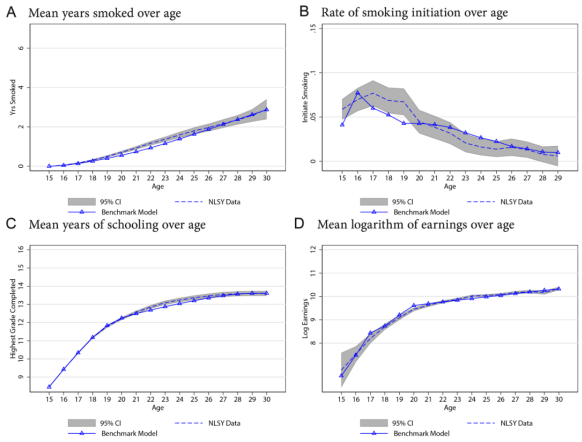

Fig. 3.—Model fit on life-cycle moments. CI = confidence interval.

Figure 3: model fit on life-cycle moments.

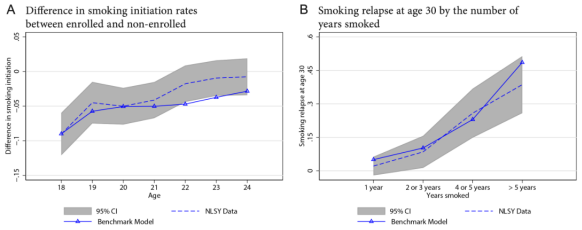

Fig. 4.—Model fit on nontargeted moments. *A*, Difference in smoking initiation rates between enrolled and nonenrolled college-aged people. *B*, Age 30 smoking relapse rate by the number of years smoked category among ever-smokers who quit smoking between ages 25 and 29. CI = confidence interval.

Figure 4: model fit on nontargeted moments.

5.6 Longevity implications: Table 4

Read.

- At age 30, nonsmokers have 5.83 more expected years of life than smokers.
- At age 30, college-educated individuals have 5.79 more expected years than non-college individuals.
- The gap between college-educated nonsmokers and non-college smokers is 6.95 years.
- At age 15, the nonsmoker-smoker expected longevity gap is 4.05 years.

Economic message. Smoking and schooling are jointly important for longevity. The gap grows from age 15 to age 30 as individuals accumulate education and addiction stock.

TABLE 4
Average Expected Longevity Difference according to Smoking and Education

	Smoking (1)	Education (2)	Smoking and Education (3)
Age 30	5.83	5.79	6.95
	(.02)	(.02)	(.02)
Age 15	4.05		
	(.08)		

Note.—Column 1 reports the difference in average expected longevity between current nonsmokers and smokers. Column 2 reports the difference in average expected longevity between individuals with and without a college education. Column 3 reports the difference in average expected longevity between individuals with a college degree who do not smoke and individuals without a college degree who smoke. Expected longevity differences by education are not calculated at age 15, because age 15 individuals have not yet graduated from high school. Standard errors are shown in parentheses.

6 Policy simulations and causal counterfactuals

6.1 Education to smoking: one extra year of schooling

- The counterfactual increases each individual's schooling by one extra year at age 20.
- Choices from ages 15 to 19 are kept identical to the benchmark.
- From age 20 onward, individuals reoptimize.
- The effect at each later age is the difference between counterfactual and benchmark smoking rates.

Read.

- Figure 5A shows that one extra year of schooling reduces the smoking rate by about 3 percentage points at age 20.
- By age 30, the smoking-rate reduction reaches about 7 percentage points.
- The effect accumulates because some people who would otherwise start smoking after age 20 never start in the counterfactual.
- Figure 5B shows heterogeneity by cognitive and noncognitive ability.

Economic message. Education is not only a labor-market investment. It changes health behavior by lowering smoking utility and increasing the value of the future.

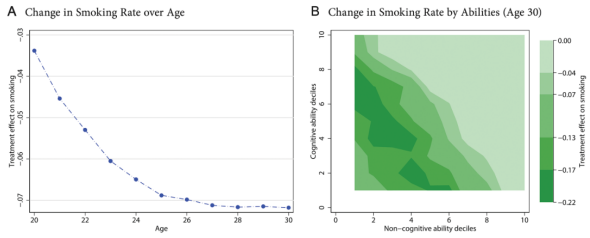

Fig. 5.—Causal effects of education on smoking rate.

TABLE 5
Effects on Quitting Smoking in the Counterfactual Experiments

	Increasing Education (1)	Revenue-Neutral Excise Tax (25%) (2)
Cognitive ability	.022***	.080***
	(.007)	(.004)
Noncognitive ability	.345***	.169***
	(.013)	(.008)
Education control	Yes	Yes
Observations	11,786	11,786
Pseudo R^2	.304	.081

Note.—Probit analysis of quitting smoking (marginal effects), among age 30 smokers in the benchmark model. The dependent variable is an indicator variable for not smoking at age 30 in the corresponding counterfactual simulation. The probit analysis also controls for years of schooling in the initial period of the corresponding counterfactual experiment. Marginal effects are evaluated at the sample means; standard errors are shown in parentheses.

*** $p < .01$.

6.2 Who quits when education increases? Table 5

Read.

- Table 5 studies benchmark age-30 smokers and asks who quits in the counterfactuals.
- In the education counterfactual, cognitive ability has a small positive marginal effect on quitting, 0.022.
- Noncognitive ability has a much larger marginal effect, 0.345.
- Since the standard deviation of noncognitive ability is about 0.51, a one standard deviation increase translates into an effect of roughly 18 percentage points.
- Under the revenue-neutral 25% tax policy, both cognitive and noncognitive ability predict quitting, with marginal effects of 0.080 and 0.169.

Economic message. Noncognitive skills, interpreted broadly as self-control and related traits, are central to the behavioral response. Education works especially strongly when combined with the ability to resist addiction.

6.3 Smoking to education: eliminating smoking

- To measure the negative effect of smoking on education, the paper removes smoking from the choice set.
- Agents then reoptimize schooling, work, consumption, and savings choices.
- This counterfactual should be interpreted as a structural causal experiment, not as a literal policy recommendation.

Read.

- Table 6 panel A shows that among benchmark smokers, college attendance rises from 0.364 to 0.415 when smoking is eliminated.
- In the whole population, college attendance rises from 0.674 to 0.695, an increase of about 2 percentage points.
- Among benchmark noncollege smokers, the no-smoking counterfactual raises college attendance from 0 to 0.088.
- Figure 6 shows that the effects are concentrated among youth with median cognitive ability and low noncognitive ability.

Economic message. Smoking reduces education most for the marginal college group: youth academically prepared enough to attend college but vulnerable to low self-control and addiction.

TABLE 6
Effects on College Attendance in Counterfactual Experiments

	Benchmark (1)	No Smoking (2)	Revenue-Neutral Excise Tax (25%) (3)
	A. College Attendance Rates in Different Simulations		
(1) Benchmark noncollege smokers	.000	.088	.089
(2) Benchmark smokers	.364	.415	.417
(3) All youth	.674	.695	.705
	B. College Attendance Regressions		
Cognitive ability	.370***	.338***	.309***
	(.004)	(.004)	(.004)
Noncognitive ability	.315***	.230***	.268***
	(.005)	(.004)	(.004)
Parents are 4-year-college graduates	.576***	.545***	.504***
	(.008)	(.007)	(.007)
Observations	48,150	48,150	48,150
Pseudo R^2	.449	.434	.448

Note.—Panel A reports the college attendance rates in different simulations. Panel B reports the probit regression estimates of the likelihood of college attendance in two different simulations. The dependent variable is an indicator variable for college attendance before age 30. Standard errors are shown in parentheses.
*** $p < .01$.

Table 6: effects on college attendance.

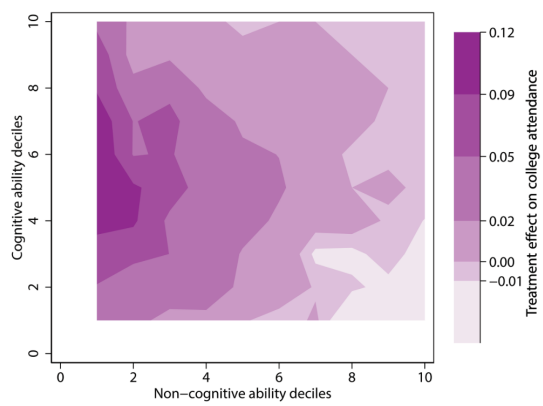

Fig. 6.—Causal effect of eliminating smoking on college attendance rate by ability deciles. Plotted are changes in college attendance rates before age 30 between the counterfactual simulation where smoking is eliminated and the benchmark simulation.

Figure 6: no-smoking effects by ability decile.

6.4 Corrective excise taxes

- The paper studies additional cigarette excise taxes from 0% to 200%.
- In the main tax counterfactual, revenue is redistributed as a lump-sum per-period tuition subsidy during schooling ages.
- The tax raises cigarette costs and the tuition subsidy lowers school costs.
- The policy is revenue neutral in the model.

Read.

- Figure 7A shows that college attendance among benchmark ever-smokers rises with the additional tax rate.
- At a 25% additional tax, college attendance reaches about 42% among would-be smokers, close to the no-smoking counterfactual.
- Figure 7B shows that the age-30 smoking rate falls with the tax rate.
- At a 25% additional tax, the age-30 smoking rate falls from 24.48 percentage points to 22.66 percentage points.
- The implied daily smoking participation elasticity is about -0.30.

Economic message. A moderate excise tax can produce human-capital gains, not merely smoking reduction. The tuition rebate matters because it channels tax revenue into the schooling margin.

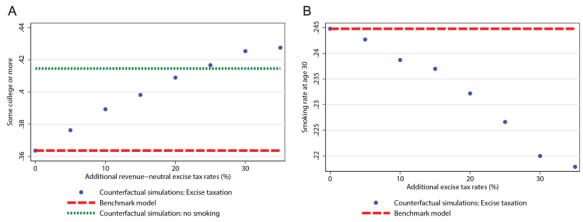

FIG. 7.—Effects of corrective excise taxes. *A*, Rate of college attendance by age 30 under different counterfactual simulations among youth who have ever smoked by age 30 in the benchmark model (i.e., benchmark model ever-smokers). *B*, Rates of daily smoking participation at age 30 under different corrective excise tax rates. The red line indicates the values in the benchmark model where there is no additional excise tax.

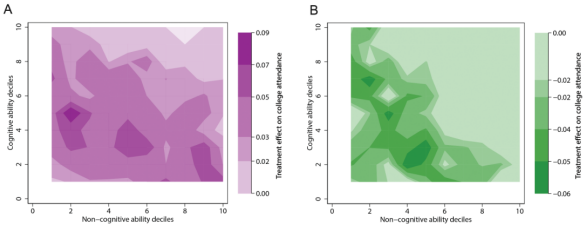

FIG. 8.—Causal effects of excise tax by abilities (25% additional tax). Plotted are changes in college attendance rate and changes in age 30 smoking rate under a 25% additional excise tax rate.

6.5 Distributional effects of the tax

Read.

- Figure 8A shows that the college-attendance gains from the 25% tax are heterogeneous by ability.
- The education effect is largest around the marginal college region, where cognitive ability is not too low but noncognitive ability is weak.
- Figure 8B shows that smoking-rate reductions also vary by ability.
- The smoking response is concentrated in the lower-triangular ability region: relatively vulnerable youth who are likely to smoke but still responsive to incentives.

Economic message. Uniform cigarette taxes have nonuniform effects. Their educational and smoking effects depend on initial endowments.

6.6 Alternative transfer schemes

Read.

- Figure 9 compares three ways to redistribute revenue from a 25% excise tax: tuition subsidy, subsidy to at-risk low-ability youth, and subsidy to nonsmokers.
- Tuition subsidies produce the largest college-attendance increase because they directly lower the cost of schooling.

- Subsidizing nonsmokers produces the largest smoking reduction because smokers pay the tax and also forgo the subsidy.
- Subsidizing low-ability youth is least effective because unconditional cash can finance both cigarettes and college.

Economic message. The transfer design changes the policy target. Tuition subsidies target education; nonsmoker subsidies target abstinence; unconditional transfers to at-risk groups can have unintended effects.

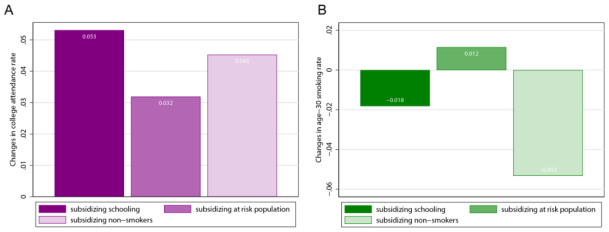

Fig. 9.—Causal effects under alternative transfer schemes with 25% additional excise tax. Plotted are changes in college attendance rate and changes in age 30 smoking rate under 25% additional excise tax rate with three different transfer schemes.

7 Limitations and interpretation

7.1 Structural assumptions

- The paper relies on a structural dynamic model, so counterfactuals depend on functional forms, state variables, and shock assumptions.
- The model assumes forward-looking agents who understand the dynamic consequences of smoking and schooling choices.
- It assumes cigarette smoking is the relevant addictive choice; other risky behaviors are not jointly modeled.

Interpretation. The model provides a coherent causal mechanism, but the results are only as credible as the model structure and the targeted moments used to discipline it.

7.2 Peer effects

- Peer effects are not directly observed in the NLSY97 data.
- The model captures peer-related variation indirectly through primitives such as cognitive ability, noncognitive ability, parental education, and age.
- The authors note that if endogenous peer effects were modeled directly, policy effects could be larger because smoking externalities would also change.

Interpretation. The paper is reduced form with respect to peers. It is designed to study internal dynamic mechanisms rather than peer-network policies.

7.3 Substitution into other risky behaviors

- The model does not let youth substitute from cigarettes into alcohol, marijuana, e-cigarettes, or obesity-related behaviors.
- This matters because modern youth use e-cigarettes and marijuana more than the NLSY97 cohort.
- The authors argue that for their cohort cigarettes were the most common tobacco product and substitution margins may be modest, but they also recognize this as a limitation.

Interpretation. The analysis is most directly about the NLSY97 cigarette cohort. The framework could be extended to newer addictive goods, but the empirical results should not be mechanically transferred without new data.

7.4 Policy interpretation

- The no-smoking counterfactual is not a practical policy but a causal decomposition.
- The 25% excise-tax counterfactual is closer to policy, especially when paired with tuition subsidies.
- The welfare interpretation is not simply "smokers are happier with taxes." The authors emphasize human-capital effects and do not claim that all smokers benefit in utility terms.

Interpretation. The main policy claim is that anti-smoking policy has an education benefit. The paper's most concrete policy implication is that cigarette taxation can be combined with schooling subsidies to target human capital.

8 Tables and figures

8.1 Figure 1, Table 1, and Figure 2: empirical starting point

Read.

- Figure 1 establishes the raw negative relationship between smoking and education in both directions.
- Table 1 quantifies the raw gaps: college-educated men smoke less, and early smokers obtain less education.
- Figure 2 shows why the timing matters: regular smoking begins early, mostly before college choices are complete.

Economic message. The descriptive facts motivate a dynamic model. If smoking begins during schooling years, it can affect human capital before schooling decisions are finalized.

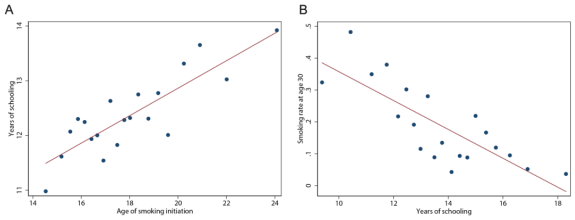

Fig. 1.—Empirical relationship between smoking and education. *A*, Binned scatterplot of the mean years of schooling as a function of age of smoking initiation among smokers. *B*, Binned scatterplot of the age 30 smoking rate as a function of years of schooling. Both plots control for cognitive and noncognitive ability average measures and parental college education. Years of schooling are measured at age 30. Source: NLSY97 males.

Figure 1. Raw smoking–education relationship.

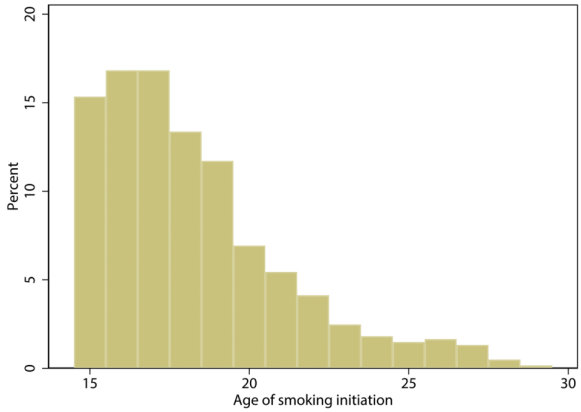

Fig. 2.—Age distribution of smoking initiation (among smokers).

Figure 2. Age distribution of smoking initiation.

TABLE 1
SUMMARY STATISTICS OF KEY VARIABLES

	(1)	(2)	(3)	(4)
	A. Youth Smoking by Education (Age 30)			
	High School or Less	College and More	Difference	p-value of Difference
Smoking	.30	.10	−.20	.00
Years smoked	4.88	1.64	−3.24	.00
Never smoked	.39	.70	.31	.00
	B. Age 30 Education by Youth Smoking History			
	Smoked before Age 20	Never Smoked before Age 20	Difference	p-Value of Difference
Years of schooling (age 30)	12.13	14.34	2.21	.00
College and more (age 30)	.34	.68	.34	.00
	C. Key Variables by Age (NLSY97)			
	Age 15	Age 20	Age 25	Age 30
Years smoked as of $t-1$	.00	.71	1.80	2.90
Never smoked as of $t-1$	1.00	.71	.63	.58
School enrollment	.97	.35	.09	.00
Working part-time while in school	.13	.26	.06	.00
Years of schooling	8.46	12.21	13.36	13.60
Earnings after leaving school ($)	2,488	17,244	29,957	40,570

8.2 Key model equations

Read.

- Equation (1) defines addiction capital as a persistent stock created by past smoking.
- Equation (2) makes subjective patience endogenous to addiction and ability.
- Equation (3) lets smoking and schooling utilities directly interact through addiction, education, ability, and shocks.
- Equations (4)–(5) show the resource constraints, where cigarette costs and tuition compete for limited resources.

Economic message. These equations are the channels through which smoking and schooling causally affect one another.

$$\beta(q_t, \boldsymbol{\theta}) := \underbrace{\exp(-\kappa_q q_t)}_{\text{effects of addiction}} \underbrace{\exp(-\kappa_r + \kappa_e \theta_e + \kappa_n \theta_n)}_{\text{effects of individual traits}}, \tag{2}$$

arameter $\kappa_q \geq 0$ captures the impact of addiction capital on iscount factors and hence on time discount rates. As addic-

Subjective discounting.

$$\begin{aligned} U(c_t, \boldsymbol{d}_t; q_t, e_t, \boldsymbol{\theta}, \boldsymbol{X}_t, \boldsymbol{\epsilon}_t) &= \frac{c_t^{1-\gamma} - 1}{1-\gamma} + \underbrace{u_q(d_{q,t}; q_t, e_t, \boldsymbol{\theta}, \epsilon_{q,t})}_{\text{smoking utility}} + \underbrace{u_e(d_{e,t}, d_{p,t}; q_t, \boldsymbol{X}_t, \boldsymbol{\theta}, \epsilon_{e,t})}_{\text{schooling utility}}, \\ u_q(\cdot) &= d_{q,t} \cdot (\nu_e e_t + \boldsymbol{\nu_\theta \theta} + \nu_0 + \epsilon_{q,t}) + \nu_1 q_t + \nu_q q_t d_{q,t} + \nu_{q^2} q_t^2 \\ &= d_{q,t} \cdot (\nu_e e_t + \boldsymbol{\nu_\theta \theta} + \nu_0 + \epsilon_{q,t}) + \underbrace{(\nu_1 + \nu_q) q_t d_{q,t}}_{\text{addictive nature of smoking}} \\ &\quad + \nu_1 q_t (1 - d_{q,t}) + \underbrace{\nu_{q^2} q_t^2}_{\text{negative health effect}}, \\ u_e(\cdot) &= d_{e,t} \cdot (\xi_q q_t + \xi_p d_{p,t} + \boldsymbol{\xi_\theta \theta} + \boldsymbol{\xi_e X}_t + \boldsymbol{\epsilon}_{e,t}), \end{aligned} \tag{3}$$

Flow utility.

$$V_t(\Omega_t) = \max_{\{\boldsymbol{d}_t, c_t, s_{t+1}\}} \{U(c_t, \boldsymbol{d}_t; q_t, e_t, \boldsymbol{\theta}, \boldsymbol{X}_t, \boldsymbol{\epsilon}_t) + \beta(q_t, \boldsymbol{\theta})\pi_t(d_{q,t}, q_t, d_{e,t} + e_t) \cdot \mathbb{E}_t[V_{t+1}(\Omega_{t+1})|_{\Omega_t}]\}.$$

After age 64, remaining lifetime utility becomes

$$V_{65}(\Omega_{65}) = \max_{\{c_t\}} \sum_{t=65}^{95} \left\{ \left(\prod_{j=65}^{t-1} \pi_j(d_{q,64}, q_j, e_{64})\beta(q_j, \boldsymbol{\theta}) \right) \left(\frac{c_t^{1-\gamma} - 1}{1-\gamma} + u_q(d_{q,64}; q_t, e_{64}, \boldsymbol{\theta}, 0) \right) \right\}.$$

The individual optimization is subject to the budget constraint, the borrowing constraint, and the law of motion of addiction stock q (eq. [1]). An individual discounts the future using his endogenous survival probability $\pi_t(\cdot)$ and subjective discount factor $\beta(\cdot)$ (eq. [2]).

2. Budget Constraint and Borrowing Constraint

An individual's budget constraint and borrowing constraint are, respectively,[9]

$$\underbrace{s_{t+1} - s_t \cdot (1 + r(s_t))}_{\text{savings in period } t} = \underbrace{Y_t + T_t + G_t}_{\text{earnings, transfers, and subsidies}} - \underbrace{(c_t + p_t \cdot d_{q,t} + m_{e_{t+1}} \cdot d_{e,t})}_{\text{consumption, addiction, and education costs}} \quad (4)$$

and

$$s_{t+1} \geq \underline{S}_{t+1}(d_{q,t}, q_t, e_{t+1}, \theta), \quad (5)$$

Dynamic objective, budget, and borrowing.

$$Z_{k,j}^* = \mu_{z,k,j} + \alpha_{z,k,j}\theta_k + \boldsymbol{x}_z'\boldsymbol{\beta}_{z,k,j} + \epsilon_{z,k,j}, \quad (8)$$

$$Z_{k,j} = \begin{cases} Z_{k,j}^* & \text{if } Z_{k,j} \text{ is continuous,} \\ \mathbf{1}(Z_{k,j}^* > 0) & \text{if } Z_{k,j} \text{ is an indicator variable,} \\ \mathbf{1}(Z_{k,j}^* > 0) + \mathbf{1}(Z_{k,j}^* > \text{cutoff}_{k,j}) & \text{if } Z_{k,j} \text{ takes values } \{0, 1, 2\}, \end{cases} \quad (9)$$

where $\boldsymbol{x}_z$ is a vector of individual control variables observed in the initial survey year 1997, including parental education, youth age in 1997, and youth schooling level in 1997. The measurement errors are assumed to be independently distributed across equations and over time. We also assume that the measurement errors follow normal distributions: $\epsilon_{z,k,j} \sim N(0, \sigma_{z,k,j}^2)$ for $k \in \{c, n\}$ and $j \in \{1, \ldots, J_k\}$.

We assume that the unobserved endowments in cognitive and noncognitive abilities are jointly normally distributed and independent of all shocks specified in the model $\boldsymbol{\epsilon}_t$ at a point in time and over time. These skill endowments are either inherited from parents or formed as a result of parental investment, due to parents' knowledge, preferences, or financial resources. The means of initial endowments may differ depending on parental education e_{pr}—that is, $\mu_k(e_{\text{pr}}) = \mu_{k,0} + \mu_{k,1}e_{\text{pr}}$, for $k = c, n$. In particular, the ex ante joint distribution of unobserved cognitive and noncognitive abilities is assumed to be

$$\begin{pmatrix} \theta_c \\ \theta_n \end{pmatrix} \sim N\left(\begin{pmatrix} \mu_c(e_{\text{pr}}) \\ \mu_n(e_{\text{pr}}) \end{pmatrix}, \begin{pmatrix} \sigma_c^2 & \sigma_{c,n} \\ \sigma_{c,n} & \sigma_n^2 \end{pmatrix} \right), \quad (10)$$

Measurement system for ability.

8.3 Estimation and fit

Read.

- Table 2 reports parameters that govern addiction, discounting, and utility.
- Table 3 shows that the model matches key regression moments linking smoking, schooling, prices, ability, and net worth.
- Figures 3–4 show that the model fits both targeted life-cycle moments and nontargeted initiation/relapse moments.

Economic message. The model is credible for counterfactuals only if it reproduces the dynamic behavioral margins. These tables and figures are the main fit evidence.

TABLE 2
KEY PARAMETER ESTIMATES

Description	Symbol	Estimate	Standard Error
Depreciation rate of addiction stock	δ_q	.3684	.0186
Subjective discount rate terms:			
Subjective discounting rate	κ_r	.0213	.0009
Myopia effect of addiction stock	κ_q	.00046	.000024
Effect of cognitive ability	κ_c	.0033	.0008
Effect of noncognitive ability	κ_n	.0032	.0014
Addiction utility parameters:			
Addiction stock	ν_q	.0224	.0014
Addiction stock squared	ν_{q^2}	−.0017	.0012
Years of schooling −9	ν_s	−.0049	.0014
Cognitive ability	ν_c	−.00001	.0022
Noncognitive ability	ν_n	−.00785	.0024
Intercept	ν_0	.0075	.0029
SD of preference shock to smoking	σ_q	.0241	.0030
Schooling utility parameters:			
Effect of addiction on schooling utility	ξ_q	−.0011	.0043
Working part-time while in school	ξ_p	−.0141	.0026
Cognitive ability	$\xi_{\theta,c}$	.0598	.0088
Noncognitive ability	$\xi_{\theta,n}$	.0380	.0132
SD of preference shock to schooling	σ_s	.2353	.0145

NOTE.—Depreciation rate δ_q of addiction stock is from eq. (1): $q_t = (1-\delta_q)q_{t-1} + d_{q,t-1}$. Subjective discount factor terms are presented in eq. (2): $\beta(q,\theta) := \exp(-\kappa_q q)\exp(-\kappa_r + \kappa_c\theta_c + \kappa_n\theta_n)$. Utility preference terms are presented in eq. (3). In particular, smoking addiction preference terms are $u_q(\cdot) = d_{q,t}\cdot(\nu_0 + \nu_1 q_t + \nu_c e_t + \nu_\theta\theta + \epsilon_{q,t}) + \nu_2 q_t^2$. Schooling preferences terms are $u_s(\cdot) = d_{s,t}\cdot(\xi_q q_t + \xi_p d_{p,t} + \xi_\theta\theta + \xi_\tau X_t + \varepsilon_{s,t})$. The parameter estimates of ξ_τ are reported in table C.VIII and discussed in sec. IV.A.3.

Table 2. Key parameter estimates.

TABLE 3
MODEL FIT: COEFFICIENTS OF REGRESSIONS FOR SMOKING, SCHOOLING, AND NET WORTH

	Data	Model	SE of Data	| Data-Model | / SE of Data
	A. Smoking Participation 2SLS Regression			
Smoking at $t-1$ ($d_{q,t-1}$)	.5072	.6687	.1494	1.0811
Smoking at $t+1$ ($d_{q,t+1}$)	.4043	.2498	.1752	.8816
Years of schooling	−.0060	−.0075	.0063	.2403
Cognitive ability	.0025	−.0047	.0052	1.3668
Noncognitive ability	−.0029	−.0292	.0092	2.8391
Log cigarette cost (log $p_{q,t}$)	−.0126	−.0138	.0183	.0650
	B. Schooling Attendance 2SLS Regression			
Smoking at $t-1$ ($d_{q,t-1}$)	−.7376	−.4618	.4233	.6517
Cognitive ability	.1277	.1423	.0135	1.0790
Noncognitive ability	.0237	.0817	.0374	1.5543
	C. Log Net Worth OLS Regression			
Cognitive ability	.2882	.2874	.1320	.0059
Noncognitive ability	.0786	.1526	.1796	.4120
Years smoked as of $t-1$	−.0525	−.0198	.0206	1.5869

NOTE.—In panel A, the dependent variable is the indicator variable of smoking in current period ($d_{q,t}$). Past and future smoking participation decisions ($d_{q,t-1}$, $d_{q,t+1}$) are instrumented using the lags and leads of excise taxes and cigarette prices. The regression analysis is restricted to postschooling observations between ages 18 and 30. In panel B, the dependent variable is the indicator variable of schooling enrollment in the current period ($d_{s,t}$). The past smoking participation decision $d_{q,t-1}$ is instrumented using past cigarette price. The regression also controls for the indicator variables for age and education. The population of the regression analysis includes individuals aged 18–24 who were previously in school. In panel C, the dependent variable is the logarithm of net worth at age 30 for individuals with at least $100. In all three panels, cognitive ability and noncognitive ability factor scores are obtained from the first-stage estimation. Parameter estimates of the constant terms are not targeted moments and hence are not reported here.

Table 3. Model fit on auxiliary regressions.

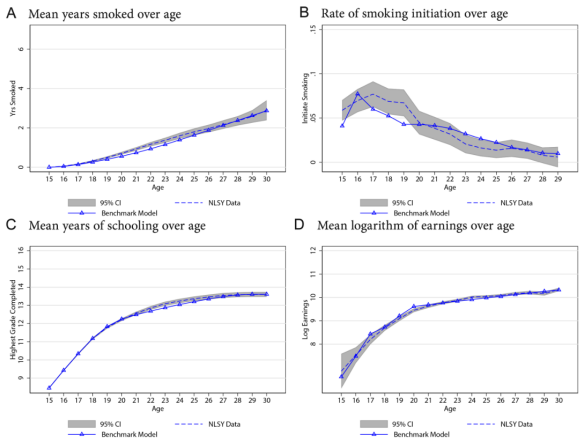

FIG. 3.—Model fit on life-cycle moments. CI = confidence interval.

Figure 3. Life-cycle fit.

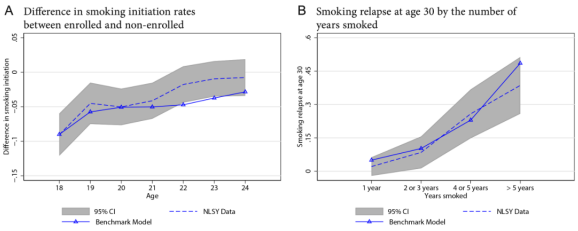

FIG. 4.—Model fit on nontargeted moments. *A*, Difference in smoking initiation rates between enrolled and nonenrolled college-aged people. *B*, Age 30 smoking relapse rate by the number of years smoked category among ever-smokers who quit smoking between ages 25 and 29. CI = confidence interval.

Figure 4. Nontargeted fit.

8.4 Counterfactuals and policy

Read.

- Figure 5 and Table 5 show that education reduces smoking and that noncognitive skills matter strongly for quitting.
- Table 6 and Figure 6 show that smoking reduces college attendance, especially for the marginal group with median cognitive and low noncognitive ability.
- Figures 7–9 show that excise taxes reduce smoking and can raise college attendance, but transfer design matters.

Economic message. The counterfactual section is the paper's policy payoff: education has health-behavior benefits, and anti-smoking policy has human-capital benefits.

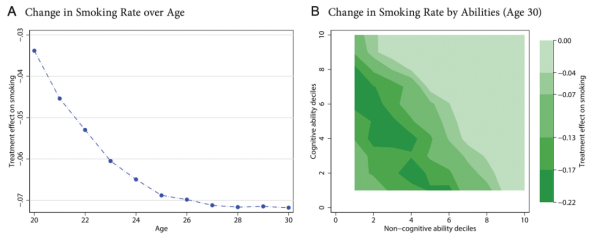

FIG. 5.—Causal effects of education on smoking rate.

Figure 5. Education reduces smoking.

TABLE 5
EFFECTS ON QUITTING SMOKING IN THE COUNTERFACTUAL EXPERIMENTS

	Increasing Education (1)	Revenue-Neutral Excise Tax (25%) (2)
Cognitive ability	.022***	.080***
	(.007)	(.004)
Noncognitive ability	.345***	.169***
	(.013)	(.008)
Education control	Yes	Yes
Observations	11,786	11,786
Pseudo R^2	.304	.081

NOTE.—Probit analysis of quitting smoking (marginal effects), among age 30 smokers in the benchmark model. The dependent variable is an indicator variable for not smoking at age 30 in the corresponding counterfactual simulation. The probit analysis also controls for years of schooling in the initial period of the corresponding counterfactual experiment. Marginal effects are evaluated at the sample means; standard errors are shown in parentheses.

*** $p < .01$.

Table 5. Who quits in counterfactuals?

TABLE 6
EFFECTS ON COLLEGE ATTENDANCE IN COUNTERFACTUAL EXPERIMENTS

	Benchmark (1)	No Smoking (2)	Revenue-Neutral Excise Tax (25%) (3)
	A. College Attendance Rates in Different Simulations		
(1) Benchmark noncollege smokers	.000	.088	.089
(2) Benchmark smokers	.364	.415	.417
(3) All youth	.674	.695	.705
	B. College Attendance Regressions		
Cognitive ability	.370***	.338***	.309***
	(.004)	(.004)	(.004)
Noncognitive ability	.315***	.230***	.268***
	(.005)	(.004)	(.004)
Parents are 4-year-college graduates	.576***	.545***	.504***
	(.008)	(.007)	(.007)
Observations	48,150	48,150	48,150
Pseudo R^2	.449	.434	.448

NOTE.—Panel A reports the college attendance rates in different simulations. Panel B reports the probit regression estimates of the likelihood of college attendance in two different simulations. The dependent variable is an indicator variable for college attendance before age 30. Standard errors are shown in parentheses.

*** $p < .01$.

Table 6. Smoking removal and tax effects on college.

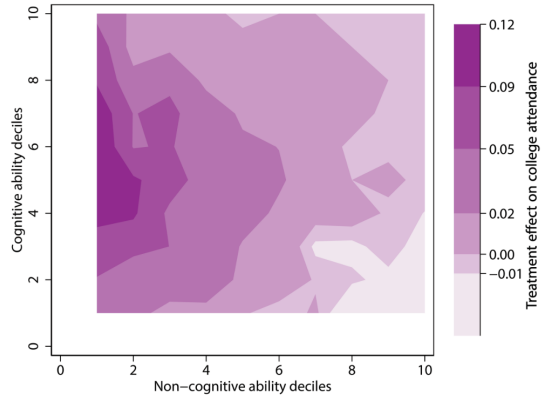

FIG. 6.—Causal effect of eliminating smoking on college attendance rate by ability deciles. Plotted are changes in college attendance rates before age 30 between the counterfactual simulation where smoking is eliminated and the benchmark simulation.

Figure 6. No-smoking effects by ability.

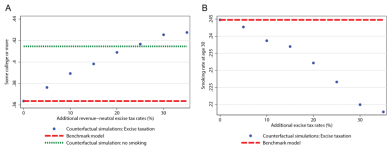

FIG. 7.—Effects of corrective excise taxes. *A*, Rate of college attendance by age 30 under different counterfactual simulations among youth who have ever smoked by age 30 in the benchmark model (i.e., benchmark model ever-smokers). *B*, Rates of daily smoking participation at age 30 under different corrective excise tax rates. The red line indicates the values in the benchmark model where there is no additional excise tax.

Figure 7. Excise tax effects.

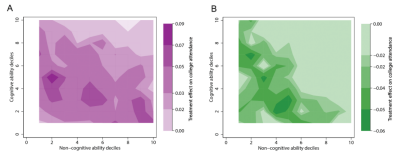

FIG. 8.—Causal effects of excise tax by abilities (25% additional tax). Plotted are changes in college attendance rate and changes in age 30 smoking rate under a 25% additional excise tax rate.

Figure 8. Tax effects by ability.

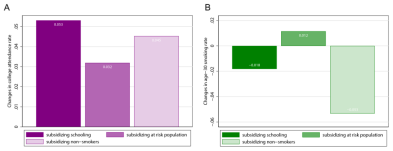

FIG. 9.—Causal effects under alternative transfer schemes with 25% additional excise tax. Plotted are changes in college attendance rate and changes in age 30 smoking rate under 25% additional excise tax rate with three different transfer schemes.

Figure 9. Alternative transfer schemes.

9 Short summary

- The paper studies reciprocal causal effects between youth cigarette addiction and education.
- The empirical raw fact is strong: more education is associated with less smoking, and early smoking is associated with less education.
- The causal problem is difficult because ability, patience, parental background, and peer-related factors affect both smoking and schooling.
- The authors estimate a dynamic life-cycle model beginning at age 15.
- Youth choose smoking, schooling, part-time work, consumption, savings, and borrowing.
- Smoking builds addiction capital, which affects future smoking utility, subjective discounting, schooling utility, survival, and borrowing.
- Education affects smoking through survival, patience, health knowledge, schooling costs, and earnings.
- NLSY97 males born in 1982 or later provide the main data.
- Cigarette prices and excise taxes provide identifying variation for smoking-related parameters.
- Latent cognitive and noncognitive abilities are estimated through a measurement system.
- The second-step SMM estimation uses 48,150 simulated individuals and 107 targeted moments.
- The model fits life-cycle smoking and schooling moments and nontargeted initiation/relapse moments.
- One extra year of schooling at age 20 reduces smoking by roughly 7 percentage points by age 30.
- Eliminating smoking raises college attendance by about 2 percentage points in the whole population.
- A 25% revenue-neutral excise tax with tuition rebates produces a similar education effect among would-be smokers.
- Noncognitive ability is central to quitting and policy responsiveness.
- Peer effects and substitution into other risky behaviors are not fully modeled.
- The broad takeaway is that anti-smoking policy can also be human-capital policy.

10 Conclusion

10.1 Core conclusion

Hai and Heckman show that youth cigarette addiction and education are dynamically linked. Education reduces smoking, and youth smoking reduces educational attainment. The paper's contribution is to model both directions inside a unified life-cycle framework with endogenous time preferences, addiction capital, schooling choices, borrowing constraints, latent abilities, and mortality risk.

10.2 Economic intuition

The key intuition is that smoking is not merely a contemporaneous consumption choice. It creates a stock of addiction that changes future preferences, patience, budget constraints, and survival. These changes reduce the attractiveness of education exactly when young people are making schooling decisions. Conversely, education changes incentives to smoke by increasing patience, lowering smoking utility, raising expected survival, and improving future earnings.

10.3 Policy takeaway

Policies that lower youth smoking can have benefits beyond health: they can increase college attendance. The model suggests that a 25% additional cigarette excise tax, if recycled as a tuition subsidy, can raise schooling among would-be smokers almost as much as eliminating smoking in the structural counterfactual. The effects are not uniform; the most responsive group has sufficient academic ability but weak noncognitive/self-control traits.

10.4 What to remember

The paper turns the education-health relationship into a dynamic two-way mechanism. Smoking reduces human capital formation, and human capital formation reduces smoking. The most important empirical lesson is that policies targeting risky youth health behavior can also change educational attainment and long-run inequality.